

Dr. Doc:

Which enzyme is primarily responsible for the hepatic metabolism of Theophylline?

A. CYP2D6

B. CYP3A4

C. CYP1A2

D. CYP2C9

✅ Answer: C.

💡 Explanation:

▼ Theophylline is extensively metabolized in the liver, primarily by the CYP1A2 enzyme.

Theophylline causes bronchodilation primarily through which mechanism?

A. Inhibition of Phospholipase A2

B. Inhibition of Phosphodiesterase (PDE)

C. Direct stimulation of β 2-receptors

D. Activation of the H1 receptor

✅ Answer: B.

💡 Explanation:

▼ Methylxanthines inhibit PDE, which degrades cyclic nucleotides, leading to increased concentrations of cAMP and subsequent bronchodilation.

What is the mechanism of action for Ipratropium Bromide?

- A. Inhibition of 5-lipoxygenase
- B. Blockade of muscarinic cholinergic receptors
- C. Stabilization of mast cells
- D. Activation of α 1-adrenoceptors

✅ Answer: B.

💡 Explanation:

▼ Anticholinergics like Ipratropium block muscarinic receptors in the bronchi to prevent bronchoconstriction and mucus secretion. ❌

Which electrolyte abnormality is a potential adverse effect of β 2-agonist therapy?

- A. Hyperkalemia
- B. Hypokalemia
- C. Hypermagnesemia
- D. Hyponatremia

✅ Answer: B.

💡 Explanation:

☀️ Adverse effects of β 2-agonists include tachycardia, hyperglycemia, hypomagnesemia, and hypokalemia.

What is a critical safety requirement for the use of Long-Acting Beta-2 Agonists (LABAs) in asthma?

- A. They should be used as monotherapy for intermittent asthma.
- B. They must be used in combination with an inhaled corticosteroid (ICS).
- C. They should only be used for acute bronchospasm.
- D. They have significant anti-inflammatory properties.

✅ Answer: B.

💡 Explanation:

▼ LABAs should only be used in combination with an asthma controller medication, such as an ICS; they have no anti-inflammatory effects.

Which of the following is a Short-Acting Beta-2 Agonist (SABA) used for "quick relief"?

- A. Salmeterol
- B. Formoterol
- C. Albuterol
- D. Indacaterol

✅ Answer: C.

💡 Explanation:

▼ Albuterol (Salbutamol), levalbuterol, and terbutaline are SABAs used for rapid relief of symptoms.

What is the primary mechanism of action for β 2-sympathomimetics in the treatment of asthma?

- A. Inhibition of Phosphodiesterase
- B. Blockade of Adenosine receptors
- C. Stimulation of β 2-receptors leading to increased cAMP
- D. Antagonism of Muscarinic receptors

✅ Answer: C.

💡 Explanation:

▼ β 2-agonists stimulate β 2-receptors, which increases cAMP formation in bronchial muscle cells, resulting in relaxation.

Which lung function test result is characteristic of Severe Persistent asthma?

- A. FEV1 > 80% of predicted
- B. FEV1 between 60% and 80%
- C. FEV1 < 60% of predicted
- D. Normal FEV1/FVC ratio

✅ Answer: C.

💡 Explanation:

▼ Severe Persistent asthma is categorized by an FEV1 < 60% of predicted and an FEV1/FVC ratio reduced by more than 5%.

A patient experiences asthma symptoms daily and is awakened by symptoms more than once a week but not nightly. How is their asthma severity classified?

- A. Intermittent
- B. Mild Persistent
- C. Moderate Persistent
- D. Severe Persistent

✅ Answer: C.

💡 Explanation:

▼ daily symptoms and nighttime awakenings >1 time/week (but not nightly) define Moderate Persistent asthma.

Which of the following is NOT a primary characteristic of bronchial asthma?

- A. Inflammation of the airways
- B. Bronchospasm
- C. Irreversible airway obstruction
- D. Bronchial hyper-reactivity

✅ Answer: C.

💡 Explanation:

Asthma is characterized by inflammation, hyper-reactivity, and bronchospasm, but unlike some forms of COPD, the airway obstruction in asthma is typically reversible.

Which side effect is more likely with oral nasal decongestants (like Pseudoephedrine) compared to intranasal forms?

- A. Bitter taste
- B. Rebound congestion
- C. Increased blood pressure and heart rate
- D. Oropharyngeal candidiasis

✅ Answer: C.

💡 Explanation:

Oral formulations result in increased systemic effects, such as elevated blood pressure, heart rate, and CNS excitation.

What is the typical onset of action for intranasal corticosteroids in chronic rhinitis?

- A. 5 to 30 minutes
- B. 1 to 2 hours
- C. 1 to 2 weeks for full effect

D. Immediately

✅ Answer: C.

💡 Explanation:

While the initial onset is 3–36 hours, patients with chronic rhinitis may not see significant improvement for 1 to 2 weeks.

Which of the following is a second-generation H1 receptor antagonist used for allergic rhinitis?

A. Diphenhydramine

B. Loratadine

C. Promethazine

D. Chlorpheniramine

✅ Answer: B.

💡 Explanation:

▼ Loratadine, fexofenadine, desloratadine, and cetirizine are second-generation antihistamines, which are generally preferred for rhinitis.

Why should intranasal α -adrenergic agonists (decongestants) be used for no longer than 3 days?

A. Risk of systemic hypertension

- B. Risk of rebound nasal congestion
- C. Risk of permanent loss of smell
- D. Risk of oral candidiasis

✅ Answer: B.

💡 Explanation:

▼ Prolonged use (over 3 days) of intranasal α -agonists leads to rebound nasal congestion (rhinitis medicamentosa).

Nasal congestion in allergic rhinitis is primarily caused by:

- A. Constriction of nasal blood vessels
- B. Dilation and increased permeability of nasal blood vessels
- C. Destruction of the nasal mucosa
- D. Decreased mucus production

✅ Answer: B.

💡 Explanation:

▼ Symptoms like nasal congestion are caused by the dilation and increased permeability of nasal blood vessels.

Dextromethorphan is chemically classified as a(n):

- A. Opioid analgesic
- B. NMDA receptor antagonist
- C. Muscarinic antagonist
- D. β 2-receptor agonist

 Answer: B.

 Explanation:

- ▼ Dextromethorphan is a synthetic central N-methyl D-aspartate (NMDA) receptor antagonist.